

ACE: Access to Careers & Education

1. Project Overview

ACE (Access to Careers & Education) is a first-of-its-kind web platform that unites education and career development in one integrated system.

It connects **university students, companies, students, teachers, and parents** in a single ecosystem that enables learning, assessment, tracking, and employment, all within the same platform.

Unlike traditional Learning Management Systems (LMS) or Career Portals, **ACE** bridges the gap between **academic education** and **career readiness** efficiently and intelligently.

2. Vision

To revolutionize how people learn, grow, and get hired by connecting **education** directly to **career opportunities**.

3. Mission

digital ecosystem that helps:

- **University students** connect with job opportunities and internships
- **Companies** discover young talents and fresh graduates.
- **Students** learn and track progress efficiently
- **Teachers** manage and monetize courses,
- **Parents** stay informed and engaged,

4. Problem Statement

Education and career development are disconnected.

Students finish courses without real-world direction, teachers lack tools to manage and grow efficiently, and companies struggle to identify qualified young talent.

5. Root Cause

The lack of an integrated system that connects **academic learning**, **career tracking**, and **real-time performance visibility** between students, parents, educators, and employers.

6. ACE's Solution

ACE solves this problem by creating a single platform that combines:

- A **career bridge** (CV uploads, job/internship postings and employment)
- A **smart learning system** (courses, exams, progress tracking)
- A **parental connection layer** (monitoring and feedback tools)
- A **teacher management system** (course creation, analytics, revenue tracking)

This integration ensures that education doesn't stop at learning — it leads to real opportunities.

7. System Users & Roles

ACE serves **five main types of users**, each with a unique experience and purpose within the system:

User Type	Description	Key Goal
University Student	Verified via academic email through Agora; accesses all course types, uploads CVs, and applies for internships or jobs.	Prepare for and transition into the workforce.
Company/Employer	Post job or internship opportunities, view student CVs, and evaluate candidates.	Discover and recruit young talents directly from the ACE ecosystem.
Student	Enrolls in courses, completes exams, earns rewards, and tracks progress by academic year.	Learn, grow, and achieve academic and career success.
Parent	Connects to their child's account to monitor progress, scores, and achievements.	Stay informed and engaged in their child's educational journey.
Teacher	Creates and manages online courses (recorded or live), tracks performance, and monitors revenue.	Deliver high-quality education efficiently and earn income.

8. Features (per User Type)

University Student

- Register using a verified academic email with OTP confirmation via Agora.
- Access general and advanced courses (not limited by grade level).
- Upload CV and showcase skills.
- Gemini AI career mentor integration that has every single detail about the university student information to help him briefly to start his career
- Full-featured and fully editable professional profile that includes experience, about me, goal, CV, education & certificates, and skills sections.
- Freely choosing whether to set the profile private or make it public, which shows the CGPA, academic year or fresh graduate, and more.
- Apply for jobs and internships from registered companies' posts on the website.
- All the features in the student profile, plus the university student features

Company

- Create a verified company profile.
- Gemini AI talent advisor integration that has every single detail about the company and it's data to help the company take better decisions about it's candidates and it's future employment
- Post internships, job offers, and training programs.
- Search and filter university student profiles by skills or GPA.
- View applicant CVs and contact candidates directly.
- View the number of jobs posted, applicants, considered applicants, scheduled meetings, active positions, and more

Teacher

- AI-powered identity verification via DIDIT verification.
- Create courses (Recorded or Live session).
- Adding lessons and deciding whether to be public or pay to watch lesson.
- Add exams, quizzes, and certificates.
- Track student performance and course ratings.
- Monitor revenue analytics per course (Average rating, Total profit, number of students enrolled in overall courses, number of courses) with analytics showing if the percentage is going up or down.
- Very detailed structured course data for the teacher that includes, for example: original price & price if there is a discount, course duration e.g. 20 hours, and number of lessons. And if it's live, then there are some more cardinalities to decide, like the number of available seats, start & end date, and a schedule for the weekly sessions (day, start & end time), and finally publish it immediately to be visible for students to start enrolling, or leave it private for now.

Student

- Sign up based on academic year (from أولى ابتدائي to ثالثة ثانوي).
- Enroll in free or paid courses.
- Gemini AI study mentor integration that has all the details and information about the student to help him briefly about anything related to the platform or related to studying in general and to improve is performance
- Take exams, earn badges, and compete for prizes.
- Personalized course recommendations based on grade level.
- Dashboard to track learning progress and achievements.
- Two types of course searching, the first is searching by a keyword or keywords that search in the course title or description, or by the teacher's name.
- Payment gateway through Stripe (bank cards) or through PayPal
- Chat for the course community.
- Pages for showing progress, my courses, exams, and more.
- Categorizing the courses, Live or Recorded

Parent

- AI-powered identity verification
- Link account to child's profile via email.
- Monitor scores, course participation, and prize history.
- Receive progress updates and recommendations.

9. System Architecture

ACE is designed using a **modular, service-oriented architecture** for scalability and flexibility.

Core Layers:

1. **Frontend Layer:** Built with React (Next.js), TypeScript, Tailwind, and Framer Motion for animations, providing a fast, efficient, reliable, and interactive multilingual, multi-theme user interface that relies on SSR technology (Server-Side Rendering).
2. **Backend Layer:** Implemented using Laravel for various reasons, like (data is structured and consistent, the need for strong relationships between data, data integrity matters, transactions are critical, the need for complex queries or reports), handling authentication, APIs, and RESTful API queries.
3. **Database Layer:** Uses MySQL for relational data (users, courses, payments, etc.).
4. **Authentication & Verification:** JWT-based auth system for security and reliability, plus academic/identity verification via DIDIT and academic email OTP via Agora.

Possible Integrations:

- Payment Gateways (Stripe & PayPal).
- Video Hosting (Vimeo API, AWS IVS, or Zoom SDK for live sessions) (not decided yet).
- Agora for sending OTP for special emails like the academic emails (because academic emails are closed for security reasons, so I needed a special approach to reach the academic emails)
- DIDIT as an AI-powered identity verification

10. Technologies

Layer	Suggested Technologies
Frontend	React.js / Next.js, Tailwind CSS, TypeScript
Backend	Laravel (PHP)
Database	MySQL
API	RESTful API for flexible data querying
Auth	JWT, Email OTP
Payments	Stripe / PayPal
Agora	Sending emails for the academic emails because the academic emails are closed and secured from any informal or unauthorized sources
DIDIT	As an AI-powered tool for identity verification